

SAT Math · Phase 3 Final Review

1

★ Mark for Review

ABC

In the given equation, k is a constant. The equation has exactly one real solution. What is the value of k ?

$$x^2 + (\sqrt{k-3})x + 42 = 0$$

A. 171**B.** 168**C.** 165**D.** 45

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ABC

In the given system of equations, m and b are negative constants. In the xy -plane, the graphs of the equations in the given system intersect at the point $(-5, y)$, where $y < 0$. Which expression represents the value of b ?

$$\begin{aligned}x^2 + y^2 &= 36 \\ y &= mx + \frac{b}{4}\end{aligned}$$

A. $-\frac{5m}{4} + \frac{\sqrt{11}}{4}$

B. $\frac{5m}{4} - \frac{\sqrt{11}}{4}$

C. $-20m + 4\sqrt{11}$

D. $20m - 4\sqrt{11}$

3

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ABC

A biologist mixed a solution that is 0.3% sodium chloride by mass with a solution that is 0.15% sodium chloride by mass to obtain a new solution, which has a mass of 80 grams and contains 0.21 grams of sodium chloride. How many grams of 0.3% sodium chloride solution did the biologist use?

A. 0.14**B.** 20**C.** 60**D.** 79.86

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ABC

The function v is defined by the equation

$$v(x) = \frac{x^2 + bx + c}{(x + 5)(x - 19)},$$

where b and c are constants. In the xy -plane, the graph of $y = v(x)$ passes through the points $(0, \frac{66}{95})$ and $(11, 0)$. If $v(q) = 0$, which of the following could be the value of q ?

A. -6 **B.** -5 **C.** 5 **D.** 6

5

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ABC

$$\frac{x}{5} + \frac{y}{9} = \frac{47}{45}$$

An engineer connects resistors in series, where the resistors in the series have a total resistance of $\frac{47}{45}$ ohms. In this series, there are x resistors of type A, which each have a resistance of a ohms, and y resistors of type B, which each have a resistance of b ohms. The given equation represents this situation. According to this equation, what is the positive difference between the value of a and the value of b ?

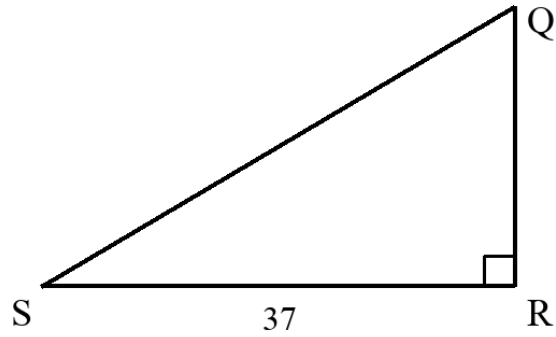
A. 47

B. 4

C. $\frac{47}{45}$ D. $\frac{4}{45}$

6

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ABC

Note: Figure not drawn to scale.

In triangle QRS shown, $QR < RS$. Which expression represents the length of $\overline{QS}$?

A. $37 \cos Q$

B. $37 \sin Q$

C. $\frac{37}{\cos Q}$

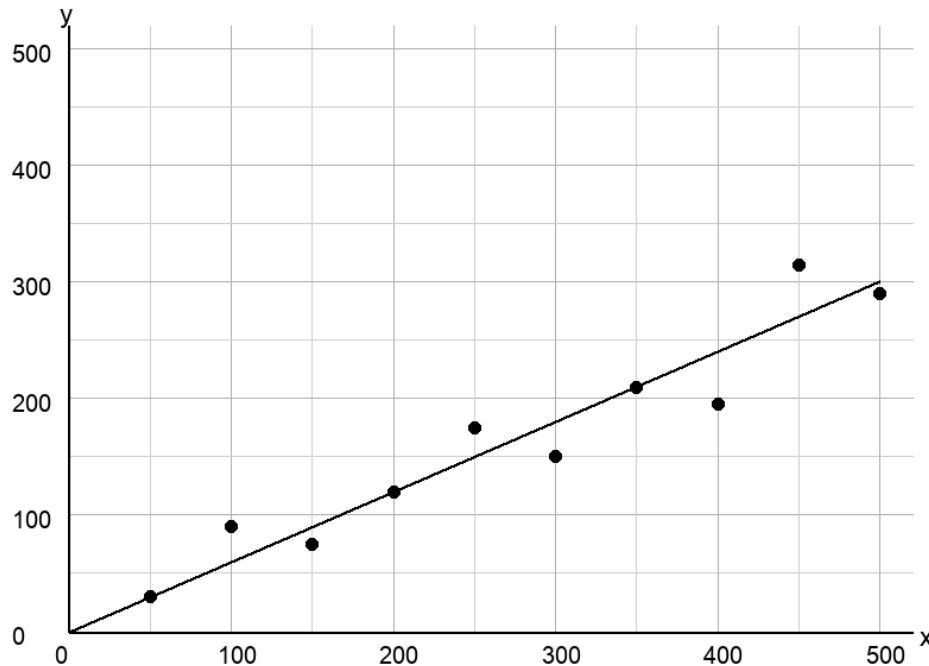
D. $\frac{37}{\sin Q}$

7

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ABC

A group of 10 gardeners recorded data on the germination rates of their tomato crop for one growing season. The scatterplot shows the relationship between the number of tomato seeds planted, x , and the number of tomato seeds that germinated, y , for each of the gardeners. A line of best fit is also shown.



Which of the following is the best interpretation of the slope of the line of best fit in this context?

- A. The number of tomato seeds planted is predicted to increase by 60 seeds every 100 days.
- B. The number of tomato seeds planted is predicted to increase by 300 seeds every 100 days.
- C. The number of tomato seeds that germinate is predicted to increase by 60 seeds for every additional 100 tomato seeds that are planted.
- D. The number of tomato seeds that germinate is predicted to increase by 300 seeds for every additional 100 tomato seeds that are planted.

8

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ABC

A rectangular banner has an area of 3,200 square inches. A copy of the banner is made in which the length and width of the original banner are each increased by 20%. What is the area of the copy, in square inches?

A. 3,220**B.** 3,240**C.** 3,840**D.** 4,608

The mass of object A is 444% of the mass of object B, and the mass of object A is 0.740% of the mass of object C. If the mass of object C is $p\%$ of the mass of object B, what is the value of p ?

10

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ABC

In 2013, Chloe earned 11% more than in 2012, and in 2014 Chloe earned 6% more than in 2013. If Chloe earned y times as much in 2012 as in 2014, which of the following is closest to the value of y ?

A. 0.5455**B.** 0.6600**C.** 0.8499**D.** 1.1766

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ABC

Point F lies on a unit circle in the xy -plane and has coordinates $(1,0)$. Point G is the center of the circle and has coordinates $(0,0)$. Point H also lies on the circle and has coordinates $(-1,y)$, where y is a constant. Which of the following could be the positive measure of angle FGH , in radians?

A. $\frac{37\pi}{2}$

B. $\frac{39\pi}{2}$

C. 34π

D. 35π